

Motivation

Training Large Language Models (LLMs) is computationally expensive, with frontier models estimating to cost upwards of \$100 Million USD during pretraining. We investigate a more efficient nanoGPT-style pretraining approach that incorporates semantic metadata. On an encyclopedia corpus, our method achieves a **3.6% reduction in bits-per-byte (BPB) loss** relative to a raw-text baseline trained on the same token budget.

Overview

Previous research shows LLM pretraining is improved by augmenting raw text with linguistic metadata such as entity or part-of-speech tags. However, efficiently producing metadata without violating causality during generation remains a challenge. We propose a new approach to embedding metadata into language modeling, characterized by three important features:

1. A much richer ensemble of tags.
2. Re-imagining metadata tags as annotated spans.
3. Avoiding data leakage by only using self-generated annotations during generation.

Findings

- We show that for a 5 million parameter nanoGPT model, we achieve a 3.6% improvement in our BPB on our test data, pretraining on the exact same amount of total tokens, without incurring the computational cost of inference on the metadata itself.
- High-quality text is a finite resource, and the current generation of LLMs already use most publicly available internet text, motivating the use of synthetic and multimodal data in pretraining. Our findings show that providing richer metadata for existing text improves LLM pretraining results, providing an alternative path to addressing data scarcity.
- Although these results are currently limited to encyclopedia data, they suggest that richer annotations may help language models use existing training data more efficiently. Future work will test whether this benefit extends to broader, general-purpose text corpora.

Data

The raw corpus was collected from the open source *Encyclopedia of World History*, featuring ~85,000 sentences of rich historical writing. This text was annotated manually with regular expressions (regex). We trained a 224 million-parameter Flan-T5 model on the initial annotated corpus for the purpose of bulk annotation of general-purpose text data. We also used it to iteratively correct our annotations on the *Encyclopedia of World History*, which was used to retrain the Flan-T5 annotator.

«[[[On] «August» 9, 378 c.e.]]»>, «the» «[«Eastern Roman»] «army»» «[[under] «the» «command»» «of» «Emperor Valens»]] «attacked» «a» «[«Gothic»] «army»» «(» «made» «[up of»] «[«Visigoths» and» «Ostrogoths»]» «that» «[» «had camped» «[[near] «the» «town»» «of» «Adrianople»]]» «(» «also» «called» «[«Hadrianopolis»]» «and» «was routed»» «)»>.
 «The» «battle»» «is» «often» «considered» «the» «(» «beginning»» «of» «the» «collapse»» «of» «the» «[«Roman Empire»]» «[[in] «the» «[» «fifth»] «century»]]»>.<

Figure 1: Annotated sentence example

Brackets

Bracket Pair	Meaning
« »	Proper Noun
« »	Adjective Phrase
« »	Possessive Personal Pronoun
...	...

Table 1: A sample of bracket definitions, 94 pairs total

Symbols

λ fraction of T spent on phase 1	C_1 phase-1 block size (annotated)
T total tokens seen during training	C_2 phase-2 block size (clean)
B batch size	N_1 phase-1 iterations
N total iterations	N_2 phase-2 iterations

Methods & Models

- **Control:** 5M parameter nanoGPT model w/ $n_layer = 6$, $n_head = 8$, $n_embd = 256$, trained on 90% of raw corpus, BPE Tokenizer w/ 8000 vocab size
- **Bracket:** Control model w/ 3 times block size to account for token count inflation from brackets so model sees same amount of subwords, control tokenizer w/ 188 brackets added as special tokens (vocab size = 8188), total tokens = $3(C_2)(B)(\frac{N}{3})$
- **Scrambled:** Identical to bracket model and tokenizer, except corpus bracket pairs are uniformly randomly replaced to ensure no semantic meaning.
- **Transfer:** Identical model and tokenizer as bracket, first 70% of training tokens are annotated text, final 30% on clean text so output has no brackets

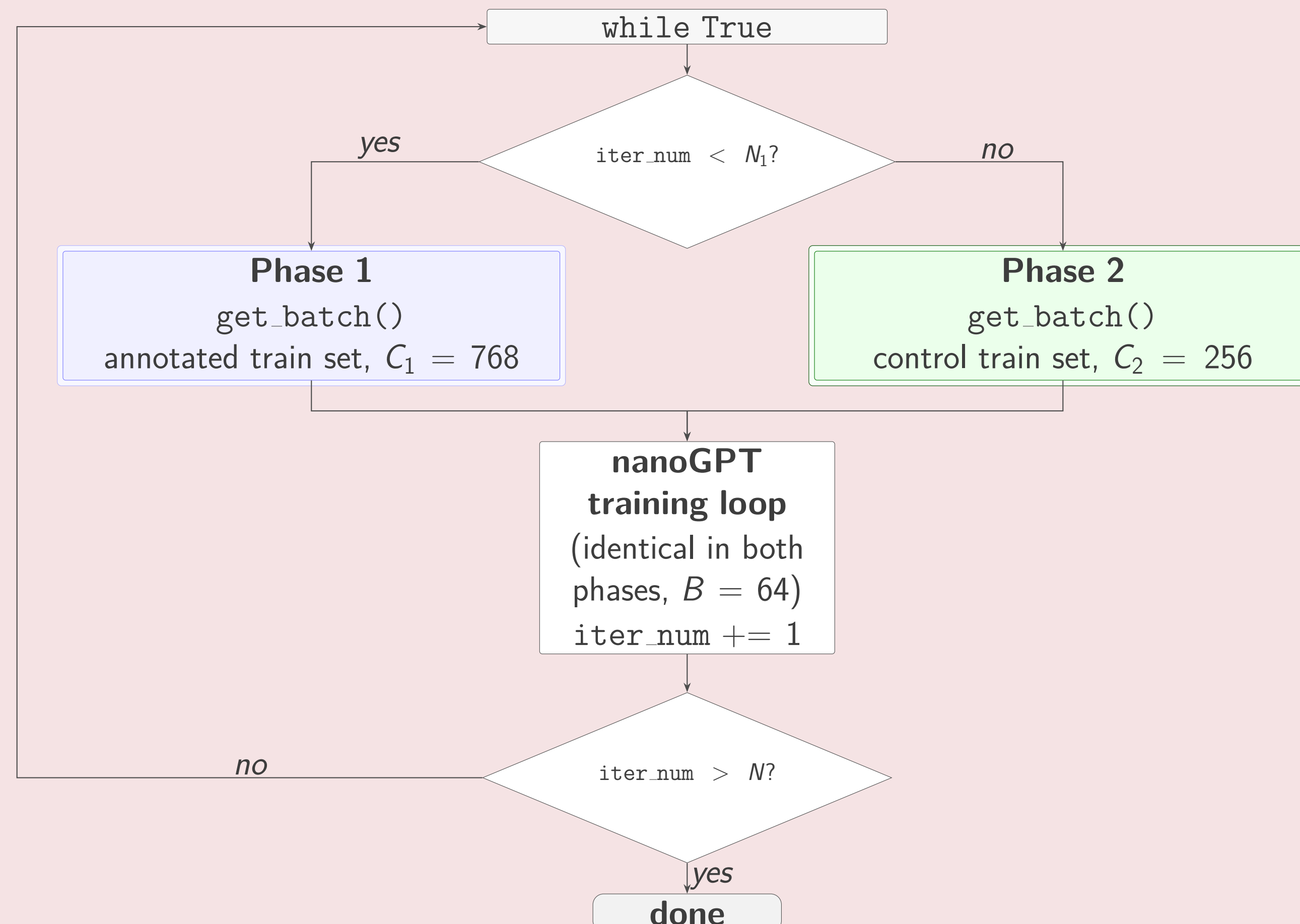
Transfer Data Loader

$$T = B(C_1N_1 + C_2N_2)$$

$$N_2 = \frac{(1 - \lambda)T}{BC_2}$$

$$N_1 = \frac{\lambda T}{BC_1}$$

$$N = N_1 + N_2$$



Results

model	Cross-Validation Split					mean	std. dev.
	A	B	C	D	E		
control	1.3995	1.3975	1.3911	1.3967	1.4009	1.3971	0.0037
bracket	1.3292	1.3304	1.3270	1.3282	1.3293	1.3288	0.0013
scrambled	2.9225	2.9146	2.9202	2.9094	2.9070	2.9147	0.0067
bracket transfer	1.3474	1.3495	1.3449	1.3475	1.3512	1.3481	0.0024

Table 2: BPB on test set across 5 train/val/test splits

Prompt	Source	Response
<i>Action was a federal agency established by President Richard Nixon on 1 July 1971. Its intention was to make the service organizations established during the 1960s operate more...</i>	Ground Truth	<i>...efficiently</i>
	Control	<i>...than nine of the National Defense Force.</i>
	Bracket	<i>...in 1971 , remaining active in</i>
	Scrambled	<i>...by “ LIA ” (2000–71)</i>
	Transfer	<i>...democratic and independently. They were</i>

Table 3: Split B: example generation

Loss Curve

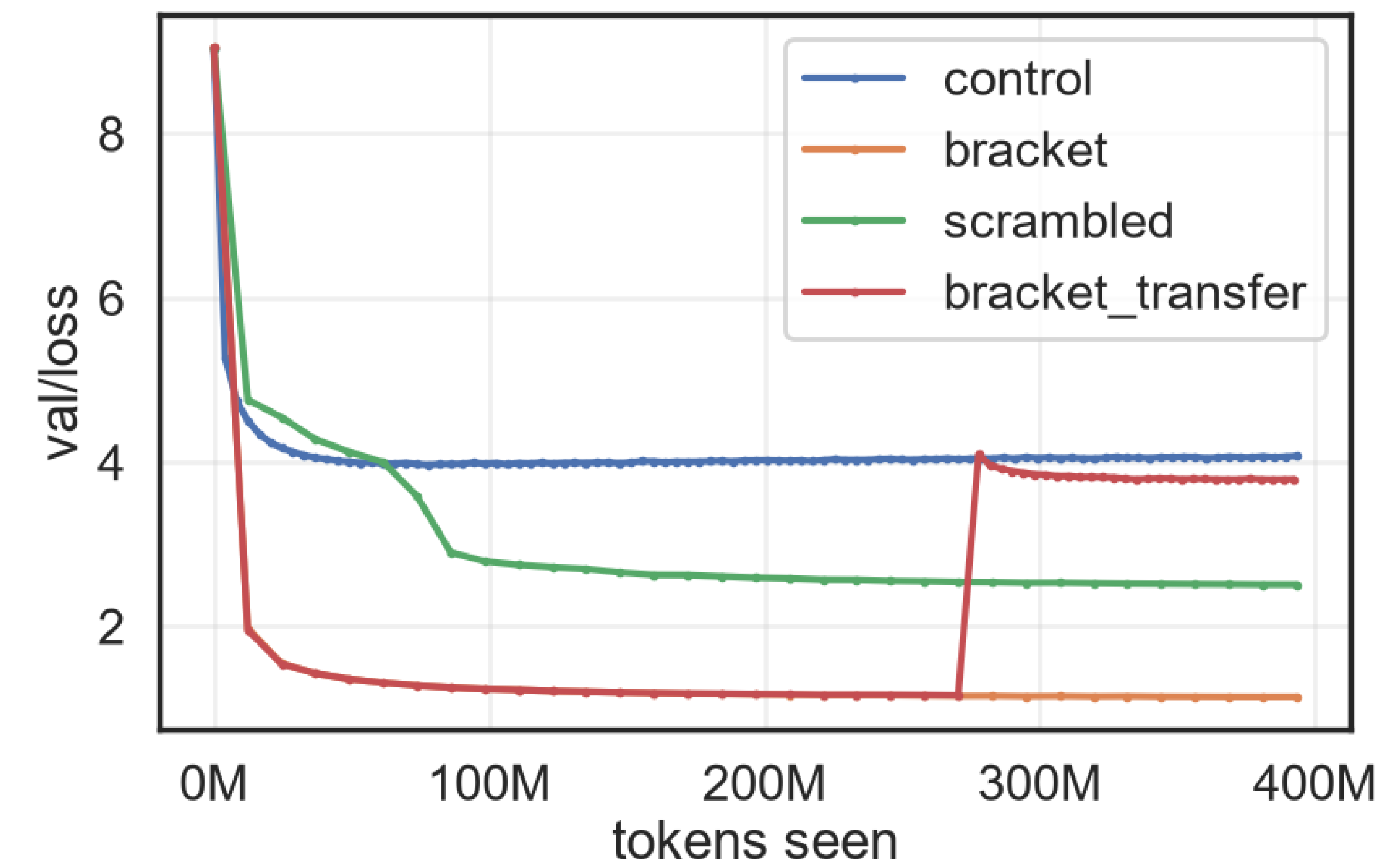


Figure 2: Split A: validation loss/tokens seen during training

Future Work

We’ve performed initial experiments on larger 124M parameter GPT-2 style models from HuggingFace with the 38GB OpenWebText corpus. We benchmark a Control and Transfer model on LAMBADA, CoQA, Arc Easy, and sst2. Initial results show notable improvements on LAMBADA and Arc Easy, specifically, and no loss of performance on the remaining two. However, a modified HuggingFace DataLoader needs to be created to switch between annotated and raw text without adding a second epoch to the transfer model, in order to ensure training is honestly comparable.

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